

M. FADIL FAIZ

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PROFESSIONAL SUMMARY

Detail-oriented Computer Engineering Technology student with a strong technical foundation in backend web development, machine learning, and hardware automation. Proven capability in building structured RESTful APIs and engineering convolutional neural networks (CNN) for complex image classification tasks. Active researcher within the campus robotics club handling embedded systems, sensor integration, and motor driver architectures. Experienced in collaborative Agile capstone project delivery and dedicated to building scalable software and intelligent autonomous systems.

EDUCATION

Politeknik Negeri Indramayu (Polindra)

2025 – Present

Bachelor of Applied Science (D4) in Computer Engineering Technology

- Current Academic Standing: Second Semester Student.
- Core Corequisites: Data Structures, Advanced Programming, Embedded Systems, and Applied Mathematics.
- Active Researcher at **UKM Robotika Polindra** (Hardware & Research Division).
- Successfully approved for a 9 SKS credit conversion from advanced external professional curriculum towards current degree tracks.

TECHNICAL SKILLS

Backend & Web:	Node.js, FastAPI, SvelteKit, JavaScript (ES6+), Python, Hapi Framework, RESTful API Design
AI & Deep Learning:	Convolutional Neural Networks (CNN), Machine Learning, Natural Language Processing (NLP), OCR, Object-Oriented Programming (OOP) in Python, Model Export & Deployment (.h5)
Databases & Tools:	PostgreSQL, MongoDB Atlas, Git, Version Control, Environment Architecture, Linux Terminal
Hardware & IoT:	ESP32, Arduino, BTS7960 High-Current Motor Drivers, Circuit Simulation (Proteus, EasyEDA), PWM Logic, Bluetooth Control Protocols

KEY PROJECTS

Web-Based AI Productivity & Automation Tool

May 2026

Collaborative Capstone Project (5-Member Team)

- Co-developed a production-ready web platform tailored for business automation and document intelligence.
- Architected core modules for automated PDF summarization and natural language processing (NLP) to parse and extract actionable items seamlessly.
- Utilized **SvelteKit** for an interactive user interface alongside **FastAPI** and **PostgreSQL** for secure database records, asynchronous processing, and OCR engine binding.

Medical Anomaly & Kidney Disease CNN Classifier

April 2026

Deep Learning Implementation Project

- Engineered a custom Convolutional Neural Network (CNN) architecture from scratch using TensorFlow/Keras to analyze medical image modalities.
- Configured a 32-neuron initial convolution layer and tuned hyper-parameters to reach a robust validation accuracy of **93%**.
- Evaluated model precision, recall rates, and confusion matrices before saving and exporting the optimized binaries to an **.h5** deployment format.

Automotive Catalog Marketplace REST API (zora-jdm)

May 2026

Backend Engineering Project

- Built a secure, clean-architecture RESTful backend system utilizing **Node.js** and Express for a Japanese Domestic Market (JDM) vehicle catalog.
- Mapped database document schemas and handled remote clusters via **MongoDB Atlas** to store comprehensive entity specifications (e.g., Nissan Silvia S15, Skyline R34, Mazda RX-7).
- Structured secure environment variable layers, data validation parsing, and production error-handling mechanisms.

Embedded Soccer Robot Control Systems

March – April 2026

UKM Robotika Research Project

- Conducted engineering research to construct an agile, competitive soccer robot for intercollegiate competitions.
- Programmed firmware on an **ESP32 microcontroller** to process incoming remote controller inputs via Bluetooth and handled custom multichannel PWM tracking logic.
- Designed industrial-grade wiring systems with **BTS7960 high-current motor drivers** to regulate motor acceleration, protect circuits, and deliver steady high-torque power.

CERTIFICATIONS & PROFESSIONAL TRAINING

Pijar x IBM SkillsBuild Scholarship: Core Analytics & Systems

February 2026

Pijar Mahir in collaboration with IBM SkillsBuild

- Successfully achieved the Mandatory Achievement 2 program milestone within the professional cloud and software developer learning track.
- Demonstrated mastery in advanced **Object-Oriented Programming (OOP) in Python**, data structures, and computer science fundamentals.

IDCamp x Indosat Scholarship: Certified Backend Developer

February 2026

Dicoding Indonesia

- Completed formal training specializing in building production-grade backend servers and network microservices.
- Acquired hands-on experience deploying APIs using the **Hapi Framework**, ensuring input sanitization, dynamic routing, and streamlined async request lifecycles.